Random Forest Model for Evaluating Recidivism Risk in Women

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***Abstract*—This article presents an innovative approach to assessing recidivism risk among female offenders in Ukraine. The research is based on the analysis of static and dynamic characteristics of 945 incarcerated women serving sentences in Ukrainian correctional facilities. The Random Forest machine learning method was applied to predict the risk of reoffending, demonstrating high effectiveness with 89.7% precision, 77.5% sensitivity, an F-measure of 82.6%, and an AUC of 95.0%. The study revealed that the most significant risk factors are suspended sentences in the criminal history, commission of medium-gravity crimes, and minor offenses. Surprisingly, marital status and having minor children showed minimal impact on recidivism tendencies. The findings indicate the necessity of developing gender-responsive approaches to probation and rehabilitation programs for female offenders in Ukraine with a focus on dynamic risk factors rather than demographic characteristics.**

Keywords—Random Forest model, machine learning, recidivism, female offenders, risk assessment, probation

I. INTRODUCTION

Crime in Ukraine is increasing, which can be directly linked to the impact of martial law conditions. At the end of 2024, approximately 39,000 people were in Ukrainian correctional facilities [1]. Among them, 92.7% were men and 7.3% were women. Women are significantly less represented among prisoners compared to men, due to their lower propensity for criminal behavior, especially violent crimes, as well as differences in socialization, pathways to crime, and sometimes more lenient treatment by the criminal justice system. Although the number of female prisoners is substantially lower than men, since 2000, there has been an increase in the number of women in detention facilities worldwide [2].

Ukraine, like many other countries, faces the problem of prison overcrowding. Probation is an alternative approach to the resocialization of offenders and helps partially solve this problem [3]. For committing minor or medium-gravity crimes, offenders are usually released from serving sentences in detention facilities with a probationary period or are assigned other types of alternative sanctions: probation supervision, community service, correctional labor, fines, and other restrictions. This reduces the burden on the penitentiary system. As of January 1, 2025, nearly 67,000 convicted persons are serving alternative sentences not involving imprisonment in Ukraine [4]. Probation promotes the rehabilitation of offenders, allowing them to remain in society, work, study, and participate in social and cultural activities. However, when providing such alternative opportunities to convicts, public safety must remain a priority in environments where offenders undergo resocialization [5].

The problem of recidivism remains one of the most serious threats to public safety when implementing probation programs. Individuals who repeatedly commit crimes after completing rehabilitation programs not only undermine public trust in the probation system but also directly threaten the safety of citizens. Some released offenders return to criminal activity during the first years after release. To effectively reduce recidivism rates, it is necessary to implement targeted risk assessment programs for each offender. This will allow for identifying individuals with a high risk of reoffending and applying enhanced supervision or alternative intervention measures to them [6]. The established increase in the number of women among prisoners creates a need for differentiated determination of risk factors for recidivism among this category of female offenders [7].

II. RELATED WORK

While a significant amount of scientific research has been dedicated to the problem of recidivism [8–10], the gender aspects of this phenomenon, especially regarding female offenders, remain insufficiently studied and fragmented [11]. In most cases, researchers focus on problems of alcoholism, drug addiction, and mental disorders in imprisoned women [12–15]. E. Moles-López and F. T. Ananos studied women sentenced to imprisonment in Spain to examine risk and protective factors associated with criminal penitentiary recidivism. The results of their study demonstrated a relationship between recidivism and various sociological and criminological characteristics, with formal education being identified as a protective factor [2]. E. Blaauw et al. conducted a study of recidivism and its factors among high-level Dutch female recidivists, analyzing their criminal cases after release. They found that although 43% of these women had registered contacts with justice within a year after release (predominantly for non-violent property crimes), the overall number of offenses significantly decreased after release. No demographic or psychiatric characteristics proved to be predictive of general

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recidivism [16]. Thomann et al. studied the effectiveness of gender-specific interventions aimed at reducing recidivism among female offenders. They determined that changes in society's expectations regarding juvenile justice policy and practice are needed [7]. S. Jeffries et al. conducted a comprehensive study of women's expectations and experiences regarding reintegration in Thailand, determining that women demonstrate multifaceted and intersecting needs that shaped their path to the prison system, through it, and out of it. They stated that factors of successful resocialization of female offenders differ depending on the country, cultural context, type of crime, and length of imprisonment [17]. Scientific investigations in this field demonstrate significant dependence of results on cultural context and peculiarities of legal systems in different countries, which complicates the formation of universal conclusions and recommendations. This necessitates additional research on factors of female recidivism to develop effective approaches to their rehabilitation.

III. METHODOLOGY

A. Proposed Approach

Our research presents an innovative methodical approach to assessing the propensity for committing recidivism, based on the analysis of anonymized statistical and dynamic characteristics of 945 imprisoned women who were serving sentences in correctional institutions of Ukraine as of January 1, 2022. Based on a literature review identifying significant risk factors for recidivism [2, 18], we conducted our classification using a unique set of characteristics specific to the convicted individuals, considering the specifics of the studied dataset. We additionally analyzed such factors as the type of crime committed (minor, medium gravity, particularly serious, and serious), marital status, and the number of minor children, which can become catalysts for gender differences. To predict the risk of reoffending by female offenders, the following factors were analyzed:

- Static factors – unchangeable characteristics that are important when assessing the risk of reoffending:
  - age 1 – the age at which the first imprisonment (1 – pre-18 years old – the juvenile period; 2 – 18-30 years old – personality is still developing, but legal responsibility is already complete; 3 – 30-45 years old – maximum social responsibility when motivation for offending typically decreases; 4 – aged over 45 – natural crime desistance through aging and changing priorities);
  - age 2 – the age at which the first arrest occurred (categories 1–4 same as for age 1);
  - marital status (1 – single; 2 – married);
  - education (0 – incomplete secondary education; 1 – secondary school education; 2 – technical or vocational training; 3 – incomplete high education; 4 – high school diploma);
  - type_of_employment (0 – unemployed; 1 – part-time; 2 – full-time);
  - minor children – number of minor children.
- Dynamic factors – variable factors that are important not only for risk assessment but also for developing various intervention programs in prison conditions:
  - real convictions – number of actual convictions in the person's prior criminal record;
  - suspended convictions – number of suspended convictions in the person's prior criminal record;
  - early dismissals – number of early dismissals in the person's prior criminal record;
  - minor_crimes – number of minor crimes in the person's prior criminal record;
  - crimes of medium gravity – the number of crimes of medium gravity in the person's prior criminal record;
  - particularly_serious_crimes – number of particularly serious crimes;
  - serious crimes – number of serious crimes.

For building a predictive model for assessing the risk of recidivism among women, we used one of the machine learning methods, Random Forest, which provides high accuracy, resistance to overfitting, and effective work with large multidimensional datasets.

B. Random Forest Method

Random Forest represents a collective machine learning approach that unites various decision trees to form a more robust predictive model. It operates by developing multiple decision trees using various portions of data and features and then aggregating their predictions.

For each tree, Random Forest selects a randomized portion of the learning dataset and evaluates only a random subset of features at each split point. The final prediction for regression is determined as (1) [19]:

$$f(x) = \frac{1}{T}\sum_{i=1}^{T} h_i\,(x) \tag{1}$$

where T is the number of trees, $h_i(x)$ is the prediction of the ith tree, and i is the indicator function.

The next key metrics are employed to evaluate the performance of binary classification models:

- true positive results (TPR) – occurrences where the model correctly identifies positive instances within the dataset;
- true negative results (TNR) – occurrences where the model accurately recognizes negative instances;
- false positive results (FPR) – occurrences where the model incorrectly assigns positive categorization to negative instances;
- false negative results (FNR) – occurrences where the model fails to recognize positive instances, erroneously categorizing them as negative.

Performance characteristics:

$$Accuracy = \frac{TPR + TNR}{TPR+TNR+FPR+FNR}, \tag{2}$$

$$Precision = \frac{TPR}{TPR + FPR}, \tag{3}$$

$$Recall = \frac{TPR}{TPR + FNR}, \tag{4}$$

$$F1\ score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}, \tag{5}$$

The AUC represents the model's ability to distinguish between positive and negative classes by measuring the

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likelihood of correctly ranking a randomly selected positive instance above a randomly selected negative instance.

IV. Results and Discussion

To assess the risk of recidivism in women serving sentences in Ukrainian correctional facilities, based on their statistical and dynamic characteristics selected for the study. The Random Forest model was constructed using RapidMiner Studio [20].

As a result of the analysis, it was established that 72.4% of women from the analyzed dataset did not become recidivists, while 27.6% were imprisoned for the first time. Similar statistics (71.2% vs 28.8%) are shown by results obtained in [2]. 82% of the total sample were aged 18 to 45 years old at the time the first arrest occurred. 79% of the total sample were first imprisoned at this same age. 22% of the total sample were married, and 78% were single. 13.86% had incomplete secondary education, 42.13% – secondary school education, 33.87% – technical or vocational training, 4.44% – incomplete higher education, 5.7% – higher education diploma. As to type_of_employment, in the total sample, 20.85% were unemployed, 55.45% were part-time, and 23.7% were full-time.

Table I presents model quality evaluation metrics for assessing the risk of recidivism in female prisoners.

TABLE I. Random Forest Performance

Criterion	*Value*	*Criterion*	*Value*
Accuracy	91.1%	Recall	77.5%
Classification error	8.9%	F Measure	82.6%
AUC	95.0%	Sensitivity	77.5%
Precision	89.7%	Specificity	96.4%

Source: compiled by the authors using RapidMiner Studio

A high precision value of 89.7% demonstrates that when the model predicts the positive class, it does so with reliability, minimizing false alarms and maintaining a low false positive rate. A sensitivity of 77.5% is relatively good, indicating that the model is capturing more than three-quarters of the positive cases. An F-Measure of 82.6% indicates strong overall performance, suggesting that the Random Forest model has achieved a good balance between correctly identifying positive cases while also avoiding false positives. An AUC of 95.0% means the model has outstanding discriminative power – it correctly ranks a randomly chosen positive instance higher than a randomly chosen negative instance 95% of the time

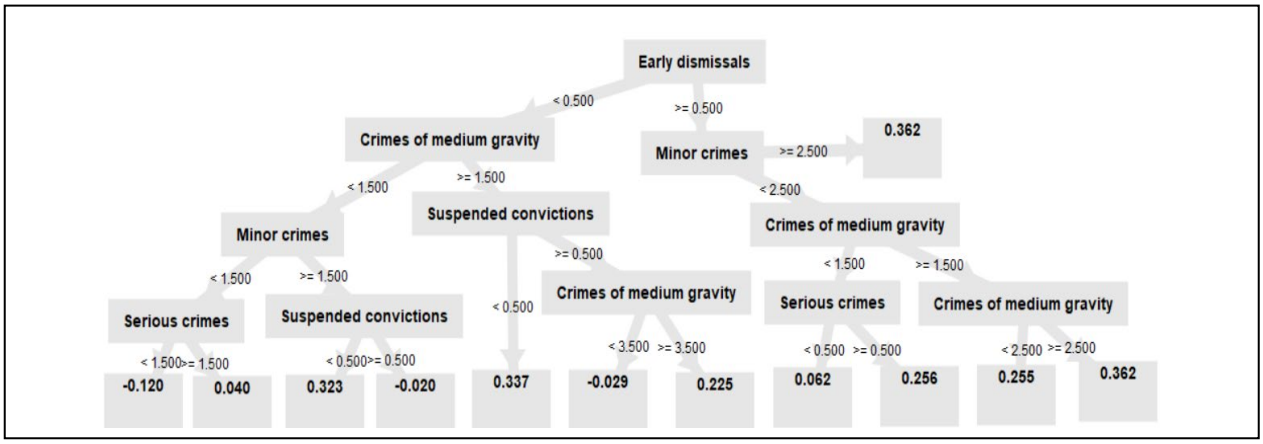

Fig. 1. Random Forest Production Model

Source: compiled by the authors using RapidMiner Studio

Fig. 1 shows the Forest Production Model, which visualizes the process of calculating the probability of recidivism by convicted women based on their statistical and dynamic characteristics.

Table II demonstrates the Random Forest model achieves excellent overall performance with high precision (above 89% for both classes) and strong recall (96.41% for class 2 – non-recidivism and 77.33% for class 1 – recidivism-prone), demonstrating reliable predictive capability.

TABLE II. Confusion Matrix for Random Forest Model

	non-recidivism	*recidivism-prone*	*class precision*
pred. non-recidivism	188	17	91.71%
pred. recidivism-prone	7	58	89.23%
class recall	96.41%	77.33%	

Table III presents the attribute weights of the created machine-learning models.

TABLE III. Weights

Attribute	*Weight*	*Attribute*	*Weight*
Suspended convictions	0.225	Type of employment	0.030
Crimes of medium gravity	0.134	Early dismissals	0.027
Minor crimes	0.072	Age1	0.025
Serious crimes	0.041	Particularly serious crimes	0.024
Age2	0.038	Marital status	0.014
Education	0.036	Minor children	0.009

The results of building the Random Forest model show that the most influential factor for the risk of recidivism in women is suspended convictions in the person's prior criminal record. This assessment is comparable with similar results obtained in our previous studies that did not take gender into account [5, 6]. This means that punishment that does not involve imprisonment is the main risk factor for female convicts. The next most influential risk factor for recidivism

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in women is the number of committed crimes of medium gravity. Therefore, women who committed this type of crime in the past most often commit repeat offenses. Women who commit minor crimes are also prone to recidivism. This result can be explained by the fact that such types of crimes are punished with a short-term punishment. Interestingly, the presence of early dismissals has a minor impact on the risk of reoffending, unlike the results we previously obtained without considering gender [5, 6]. Surprisingly, marital status and the presence of minor children have the least impact on the risk of recidivism in women. In the analyzed dataset, 52% of women are childless, 23% have one child, 14% have two children, and 11% are mothers of more than two children. The obtained results provide grounds to formulate preliminary assessments that gender is significant when evaluating the risk of recidivism in prisoners; however, dynamic factors, rather than statistical factors that determine a woman's marital status, carry the main weight.

V. CONCLUSION

This study presents a new approach to assessing the risk of recidivism among female offenders in Ukraine using the Random Forest model. Our analysis of 945 imprisoned women showed that 72.4% did not become recidivists, while 27.6% were convicted for the first time, which is consistent with similar statistics from other studies. The developed model demonstrated excellent performance with high accuracy (89.7%), good sensitivity (77.5%), strong F-measure (82.6%), and exceptional discriminatory ability (AUC 95.0%).

The study found that suspended sentences in a person's previous criminal history are the most significant risk factor for female recidivism, which is consistent with our previous gender-neutral studies. The number of medium-gravity crimes emerged as the second most influential factor, followed by minor crimes. Unexpectedly, marital status and the presence of minor children showed minimal impact on the risk of recidivism, suggesting that for female offenders, dynamic factors related to criminal history carry more weight than static demographic characteristics.

These results are important for developing gender-oriented approaches to probation and rehabilitation programs in Ukraine. By focusing interventions on women with a history of suspended sentences and medium-gravity crimes, correctional services can more effectively allocate resources to reduce recidivism rates. Additionally, the relatively low impact of family-related factors challenges traditional assumptions about women's pathways to crime and desistance.

Future research should examine the interaction between these risk factors and consider qualitative aspects of women's experiences in the criminal justice system. Developing specialized risk assessment tools specifically designed for female offenders could significantly improve the effectiveness of rehabilitation efforts and, ultimately, contribute to improving public safety and reducing prison overcrowding in Ukraine.

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